

AI Logic Branch Chain: Inventory Assessment

Logic Branch	IF (Condition / Trigger)	THEN (Action / Next Step)	Additional Context / Notes
Matrix Exception: Exotic	IF the item is categorized as "singular" or "imported exotic"...	THEN ignore 'Volume' weighting and emphasize 'Days on Hand'.	Adjusts the model for high-value items that naturally move slower but carry high margins.
Matrix Exception: Bulk	IF the item is categorized as a "large quantity" stock item...	THEN ignore 'Days on Hand' weighting and emphasize 'Volume' sold and 'Unit Cost Risk'.	Ensures high-turnover commodities aren't penalized for high stock levels.
Re-Order Trigger	IF (Current Stock + In-Transit Stock) <= (Average Daily Units Sold × Days to Arrive)...	THEN trigger stock replenishment alert and move to Financial Check (Step 3).	Mathematically ensures stock won't run out before the supplier can deliver.
Financial Approval	IF the Quick Ratio is healthy AND available cash > upcoming Capex + seasonally adjusted operational expenses...	THEN proceed to Step 4: Order Stock.	Ensures the business has adequate liquidity to survive the cash conversion cycle of the new stock.
Financial Rejection	IF the Quick Ratio indicates low liquidity OR upcoming Capex requires the available cash...	THEN halt the order OR reduce order quantities to match strictly necessary daily minimums.	Prevents inventory purchases from bankrupting operational cash flow.